



Figure 2: The iPhone UI elements

fair. Refer to Figure 1 to understand its architecture.

- Can vary in clock speed up to 700MHz or more, depending on implementation.
- ARM Intelligent Energy Manager (which is supposed to reduce power consumption by 25-50 per cent in portable devices).
- 16K/16K cache
- Features a vector floating point coprocessor (for embedded 3D-graphics)
- ARM Jazelle enabled for embedded Java execution
- SIMD, high performance integer CPU (8-stage pipeline, 675 Dhrystone, 2.1 MIPS)
- 0.45 mW/MHz power draw (with cache)

After reading the sixth point, you might think that iPhone 3G supports Java, but that is not the case. But we do have a way by which we can install Java on iPhone.

iPhone supports the OpenGL ES specification. It is arguably the best 3D API available for mobile devices, and existing ES applications can be ported very easily.

iPhone UI elements (visual look)

I know it is absurd to suddenly talk about the UI elements, but I am so impressed by the iPhone's look and feel that I thought it would be a good idea to impress you as well.

The deal is, the iPhone default interface is awesome (well, looks *do* sell!) and the SDK enables every developer to create an application with the same look and feel. So, third-party applications will essentially look integrated into the iPhone OS. You may very well give users the feel of a factory-bundled application when you develop your own app.

Hope the collage in Figure 2 gives you an idea of what I am talking about.

The OS

Now, tell me what could possibly be the best thing that could happen to an iPhone? iPhone running Linux as an operating system, right? Not quite... But it's running UNIX at least. Or actually running an open source kernel called XNU.

iPhone OS is essentially the stripped down version of the Mac OS X in almost every aspect. That means the iPhone OS:

- Runs a kernel called XNU (X is not UNIX), which is a hybrid kernel based on MACH (operating system micro-kernel developed at Carnegie Mellon University) and BSD.
- Is compatible with the Single UNIX Specification version 3 (SUSv3) and POSIX UNIX applications and utilities, including Bash, cat, etc.
- Supports kernel cache, a flat file with all drivers loaded.
- The UI abstraction layer (responsible for awesome UI elements) is powered by Cocoa Touch, which is based on the Mac OS X desktop version of Cocoa API.

There is much more to the MAC OS X operating system when we look at it from a UNIX point of view. I would like to cover it in detail, if Linux fans have no problems looking into other UNIX variants.

iPhone development: The Apple way

It's time to get started, but first the bad news. Apple does not provide an SDK for Linux. The iPhone SDK is only available for Mac OS X. However, the good news is that there is an Open Toolset that we can use on Linux, to develop for iPhone.

Historically, iPhone has been a closed device and still is.

The Apple way to develop for iPhone looks like this:

1. Join the iPhone Developer Program at developer.apple.com/iphone/program with a membership fee of \$99.
2. Download the iPhone SDK from developer.apple.com/iphone/program/download. The SDK is entirely free (of cost). But deploying on the real iPhone requires paid membership.

The iPhone SDK essentially includes a world class

IDE called Xcode. It is a suite of tools for developing software on Mac OS X, created by Apple. The Xcode suite includes a modified version of free software GNU Compiler Collection (GCC, `apple-darwin9-gcc-4.0.1` as well as `apple-darwin9-gcc-4.2.1`, with the former being the default), and supports C, C++, Fortran, Objective-C, Objective-C++, Java, AppleScript, Python and Ruby source code with a variety of programming models, including Cocoa, Carbon, and Java.

Third parties have added support for GNU Pascal, Free Pascal, Ada, C#, Perl, Haskell and D. The Xcode suite uses GDB as the back-end for its debugger. Xcode can be used to compile and debug applications for the ARM processor used in iPhone and iPod Touch.

If you ask me, Xcode is probably the best IDE available that uses GCC and GDB as the backends. In other words, it's way better than Visual Studio and it uses the GNU toolset. Xcode is available free from developer.apple.com/tools/xcode.